List of references: Ergodic thory / Dynamical systems and related fields

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Dynamical systems and Ergodic theory are broad subjects. It consists of various aspects of different behaviors and phenomena. I tried to introduce books that would be helpful for graduate students and early-career researchers. These books contain good materials to learn what has been studied in the research area. Please let me know if you are aware of any other books not included in the list.

Suggested reading for all students

1. The Princeton Companion to Mathematics, Timothy Gowers, June Barrow-Green, Imre Leader. - This book contains a great introduction to general research topics!
2. An Epsilon of Room, I: Real Analysis An Epsilon of Room, I: Real Analysis: pages from year three of a mathematical blog, T. Tao

Basic texts for Ergodic theory and Dynamical systems

1. Introduction to Dynamical Systems, G. Stuck and Michael Brin, 2002
2. A First Course in Dynamics: With a Panorama of Recent Developments, Boris Hasselblatt and Anatole Katok
3. Introduction to the Modern Theory of Dynamical Systems, Anatole Katok, 1995
4. Ergodic Theory: with a view towards Number Theory, Manfred Einsiedler and Thomas Ward.
5. Ergodic Theory, Karl E. Petersen.
6. Lectures on ergodic theory, Halmos
7. Geodesic and Horocyclic Trajectories, Françoise Dal'Bo.
8. T. Tao's blog - Ergodic theory. [Link]
9. Henk Bruin's note [pdf]
10. Ergodic Theory: A Probabilistic Approach to Dynamical Systems, Alex Blumenthal, Lai-Sang Young
11. Ergodic Theory and Differentiable Dynamics, Ricardo Mané

Ordinary Differential Equations

1. Ordinary Differential Equations and Dynamical Systems, Gerald Teschl [link]
2. Ordinary Differential Equations with Applications, Carmen Chicone
3. Differential Equations and Dynamical Systems, Lawrence Perko
4. Nonlinear dispersive equations: local and global analysis, Terrence Tao

These books are only suggested for reading based on your research interest.

Advanced topics in Ergodic theory

1. Introduction to Smooth Ergodic Theory, Luís Barreira and Yakov Pesin
2. Lecture Notes on Ergodic Theory, O. Sarig [pdf]
3. Introduction to Partially hyperbolic dynamics, Sylvain Crovisier and Rafael Potrie [pdf]
4. Ergodic Theory via Joinings, Eli Glasner
5. An Introduction to Infinite Ergodic Theory, Jon Aaronson.
6. Nilpotent Structures in Ergodic Theory, Bernard Host and Bryna Kra
7. Notes on the Multiplicative Ergodic Theorem, Simion Filip [pdf]

Interval exchange maps / Teichmüller theory

1. Introduction to Teichmüller theory and its applications to dynamics of interval exchange transformations, flows on surfaces, and billiards. Forni and Matheus [pdf]
2. Ergodic theory of interval exchange maps, Viana [pdf]
3. Translation surfaces, Arthreya and Masur
4. A Primer on Mapping Class Groups, Benson Farb and Dan Margalit

Symbolic dynamics

1. Topological and Ergodic Theory of Symbolic Dynamics, Henk Bruin [pdf]

Hyperbolic dynamics and related topics

1. Hyperbolic flows, Todd Fisher and Boris Hasselblatt
2. Zeta functions and the periodic orbit structure of hyperbolic dynamics. Parry.
3. Global Stability of Dynamical Systems, Shub
4. Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms, Robert Edward (Rufus) Bowen, Jean-René Chazottes, David Ruelle
5. Dynamical Systems and an example: Billiards, Carlangelo Liverani [pdf]

6. Chaotic Billiards - Nikolai Chernov, Roberto Markarian
7. Microlocal Analysis in Hyperbolic Dynamics and Geometry, Thibault Lefeuvre.
8. Dimension theory in dynamical systems, Ya. B. Pesin
9. Dimension Theory of Hyperbolic Flows, Luis Barreira
10. Hyperbolic Dynamics and Brownian Motion: An Introduction, Jacques Franchi, Yves Le Jan

Transfer operator methods

1. Stochastic Dynamics of Deterministic Systems, Viana [pdf]
2. Introduction to the transfer operator method, O. Sarig [pdf]
3. Transfer Operators in Hyperbolic Dynamics, Nilofar, Carlangelo Liverani, Demers
4. Statistical properties of dynamics. Introduction to the functional analytic approach, S. Galatolo [pdf]
5. Perturbation Theory for Linear Operators. Kato.
6. Positive Transfer Operators and decay of correlations, V. Baladi
7. Dynamical Zeta Functions and Dynamical Determinants for Hyperbolic Maps, V. Baladi

Homogeneous dynamics

1. Ergodic Theory and Topological Dynamics of Group Actions on Homogeneous Spaces, M. Bachir Bekka and Matthias Mayer
2. Dynamical systems on homogeneous spaces, Starkov
3. Ratner's Theorems on Unipotent Flows, Dave Witte Morris
4. Homogeneous Dynamics and Applications, Manfred Einsiedler and Thomas Ward. [Link]
5. Entropy in ergodic theory and homogeneous dynamics, E. Lindenstrauss, Manfred Einsiedler and Thomas Ward. [Link]

Rigidity and cocycles in higher rank actions

1. Rigidity in Higher Rank Abelian Group Actions, Anatole Katok and Viorel Niţică
2. Combinatorial Constructions in Ergodic Theory and Dynamics. Anatole Katok.
3. Geometry, Dynamics, and Rigidity. Amie Wilkinson

One-Dimensional Dynamics

1. One-Dimensional Dynamics, Welington de Melo &, Sebastian van Strien
2. One-Dimensional Dynamics From Poincaré to renormalization, Yiheng Dong, Marco Martens, and Liviana Palmisano

Lyapunov Exponents

1. Lectures on Lyapunov Exponents, Viana
2. Lyapunov Exponents of Linear Cocycles: Continuity via Large Deviations. Pedro Duarte, Silvius Klein

Random dynamics

1. Random Dynamical Systems, Ludwig Arnold.
2. Random Perturbations of Dynamical Systems, Yuri Kifer
3. Local Limit Theorems for Inhomogeneous Markov Chains, Dolgopyat and Sarig.
4. A Course on Large Deviations with an Introduction to Gibbs Measures, Firas Rassoul-Agha, Timo Seppalainen

Probability and Stochastic Process

1. Probability and Measure, Patrick Billingsley
2. Probability Theory and Examples, Richard Durett
3. Essentials of Stochastic Processes by Richard Durrett
4. Theory of Probability and Random Processes, Leonid Koralov, Yakov G. Sinai
5. Markov Chains and Mixing Times, David A. Levin, Yuval Peres, and Elizabeth Wilmer
6. Markov Processes, Structure and Asymptotic Behavior, Murray Rosenblatt
7. Measure Theory, Probability, and Stochastic Processes, Jean-François Le Gall
8. Brownian Motion, Martingales, and Stochastic Calculus, Jean-François Le Gall
9. Stochastic Differential Equations: An Introduction with Applications, Bernt Øksendal
10. Brownian Motion and Stochastic Calculus, Ioannis Karatzas and Steven E. Shreve
11. Continuous Martingales and Brownian Motion, Daniel Revuz & Marc Yor
12. An Introduction to the Theory of Point Processes vol I, II, D.J. Daley D and Vere-Jones.
13. Chaos, Fractals, and Noise: Stochastic Aspects of Dynamics, Andrzej Lasota and Michael C. Mackey

Random walks

1. Two-Dimensional Random Walk, Serguei Popov
2. Non-homogeneous Random Walks, Mikhail Menshikov, Serguei Popov, and Andrew Wade.
3. Principles of Random Walk, F. Spitzer
4. Random walks on reductive groups, Jean-François Quint, Yves Benoist
5. Random walks on infinite graphs and groups, Woess

Relations with Lie groups and representation

1. Unitary Representations and Unitary Duals, Manfred Einsiedler and Thomas Ward.
2. Representations of nilpotent Lie groups and their applications, Corwin and Greenleaf
3. Ergodic theory and semisimple groups, Zimmer
4. Discrete subgroups of Semisimple Lie groups, Margulis
5. Representation theory of semisimple groups, Knapp
6. Harmonic Analysis on Semi-Simple Lie Groups I/II, Warner, Garth

Applications in Number Theory

1. Continued Fractions, Khinchin
2. Metric Number Theory, G Harman
3. Metric Diophantine approximation and dynamical systems, Dmitry Kleinbock [pdf]
4. Sequences, Discrepancies and Applications, Michael Drmota and Robert F. Tichy
5. Equidistribution in Number Theory, An Introduction, Andrew Granville, Zeév Rudnick

KAM theory

1. A tutorial on KAM theory, R. Llave.
2. The KAM Story: A Friendly Introduction by H. Scott Dumas
3. KAM Theory and Semiclassical Approximations to Eigenfunctions by Vladimir F. Lazutkin
4. Weak Kam Theorem in Lagrangian Dynamics, Albert Fathi

Hamiltonian systems

1. Mathematical Methods of Classical Mechanics by V.I. Arnold
2. Symplectic Invariants and Hamiltonian Dynamics, Eduard Zehnder and Helmut Hofer

Semi-classical Analysis

1. An Introduction to Semiclassical and Microlocal Analysis by Martinez
2. Semiclassical Analysis by Maciej Zworski
3. Semiclassical Analysis by Guillemin and Sternberg

Further reading: miscellaneous

1. Fuchsian Groups, Svetlana Katok.
2. Mathematical Foundations of Information Theory by A. Ya. Khinchin
3. Geometry and Billiards, Serge Tabachnikov

Survey for the ICM Proceedings

Giovanni Forni [pdf], Corinna Ulcigrai [pdf], Simion Filip [pdf], Hee Oh [pdf], Jérôme Buzzi [pdf], Frédéric Faure [pdf] Dmitry Dolgopyat [pdf], Pierre Berger [pdf], Danijela Damjanovic [pdf], Adam Kanigowski [pdf] etc...

Collections of articles and surveys

1. Dynamics, Geometry, Number Theory: The Impact of Margulis on Modern Mathematics
2. Collected Works of Anatole Katok, (in 2 Volumes)
3. Modern Dynamical Systems and Applications. 2004